Lesson for the course “Human-Computer Interaction”

Evaluation

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Why, what, where and when to evaluate

- Iterative design and evaluation is a **continuous** process
- **Why**: to assess users' requirements and check that users can use the product and they like it
- **What**: a conceptual model, early prototypes of a new system and later, more complete prototypes and to compare them with competitors' products
- **Where**: in natural, in-the wild, and laboratory settings
- **When**: throughout design and even after (finished products can be evaluated to collect information to inform new products)

Importance of evaluation

- *"Iterative design, with its repeating cycle of design and testing, is the **only validated methodology in existence that will consistently produce successful results**. If you don't have user-testing as an integral part of your design process you are going to throw buckets of money down the drain." - Bruce Tognazzini*

Evaluation settings

- **Controlled settings** that directly involve users
  - E.g., usability labs and research labs

- **Natural settings** involving users
  - E.g., online communities and products that are used in public places
  - Often there is little or no control over what users do, especially in in-the-wild settings

Participants' rights and getting their consent

- Participants need to be told why the evaluation is being done, what they will be asked to do and informed about their rights
- **Informed consent forms** provide this information and act as a contract between participants and researchers
- **Remember to provide the information sheet and consent form during the evaluation phases of your exam project**
  - E.g., you may specify that you are going to record a video of the user interacting with the prototype

Things to consider when defining the evaluation and interpreting the results data

- **Reliability**: does the method produce the same results on separate occasions?
- **Validity**: does the method measure what it is intended to measure?
- **Ecological validity**: does the environment of the evaluation distort the results?
- **Biases**: Are there biases that distort the results?
- **Scope**: How generalizable are the results?

Usability testing

- The main benefit and purpose of usability testing is to identify usability problems with a design **as early as possible**, so they can be fixed before the design is implemented or mass produced
- As such, usability testing is often conducted on prototypes rather than finished products, with different levels of fidelity (i.e., detail) depending on the development phase

Usability testing

- Is conducted in controlled settings
- In a typical usability test, a test moderator gives participants a series of **tasks** that they must perform with the design
- The tasks represent actions that an end user would typically carry out with the finished product
- During the test, the moderator **observes** each participant's actions, often also **recording** the test session on video (in some cases also including eye tracking) and **saving log data** (e.g., key presses, clicks)

Quantitative performance measures

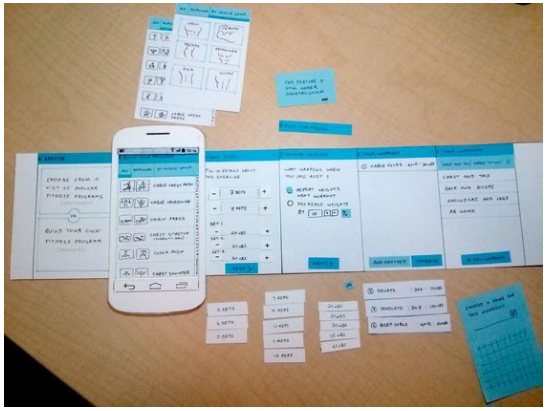
- Number of users successfully completing the task
- Time to complete task
- Time to complete task after time away from task
- Number and type of errors per task
- Number of errors per unit of time
- Number of navigations to online help or manuals
- Number of users making a particular type of error
- User satisfaction is evaluated using questionnaires and interviews

Usability testing

- After analyzing the results of a usability test, the moderator **reports on several points of interest** that arose
- These include **issues** such as the aspects of the design that **caused** problems and the **severity** of these problems, as well as places in the design that the participants particularly **liked**
- Recognizing this potential to highlight difficulties and strong points in a design's early versions is a vital part of a designer's thought process

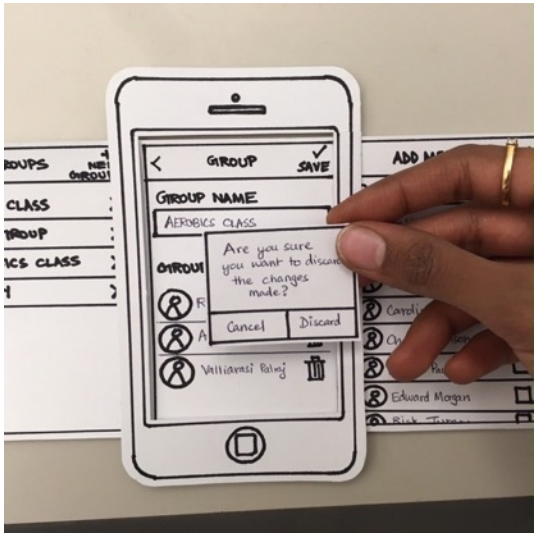
Example: usability testing of paper prototype

- Give users a task and find a way to represent the interaction with the device
  - E.g., use post-it or other ways to change the screen layout according to the choices of the tester







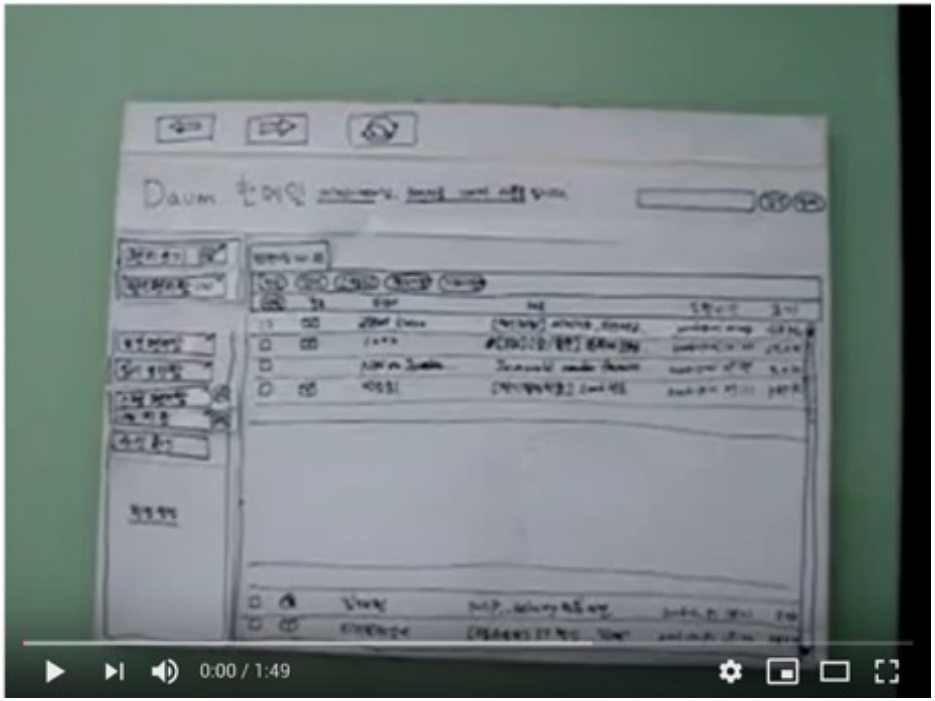


Evaluation of paper prototypes

1. **Have a list of tasks that the users need to perform**

2. Two experimenters are involved
   - The first is the **"facilitator"** that gives the user the tasks
   - The second person plays the role of the **"computer"**, i.e., he/she changes the screens
     - The "computer" **needs to rehearse** to make sure he/she is familiar with the screens before the testing session

Video of usability testing of paper prototype



- Paper prototype of webmail service
- https://youtu.be/GrV2SZuRPv0
- In this video the simple observation technique is used

Simple observation technique

- The user is given a task, the evaluator just watches the user and takes notes of the user's behaviour
- Video recording can be useful for reporting data
- Problem: this kind of evaluation **does not provide insights of the user's decision process or doubts**

- It is often better to use the think aloud technique

Think Aloud technique

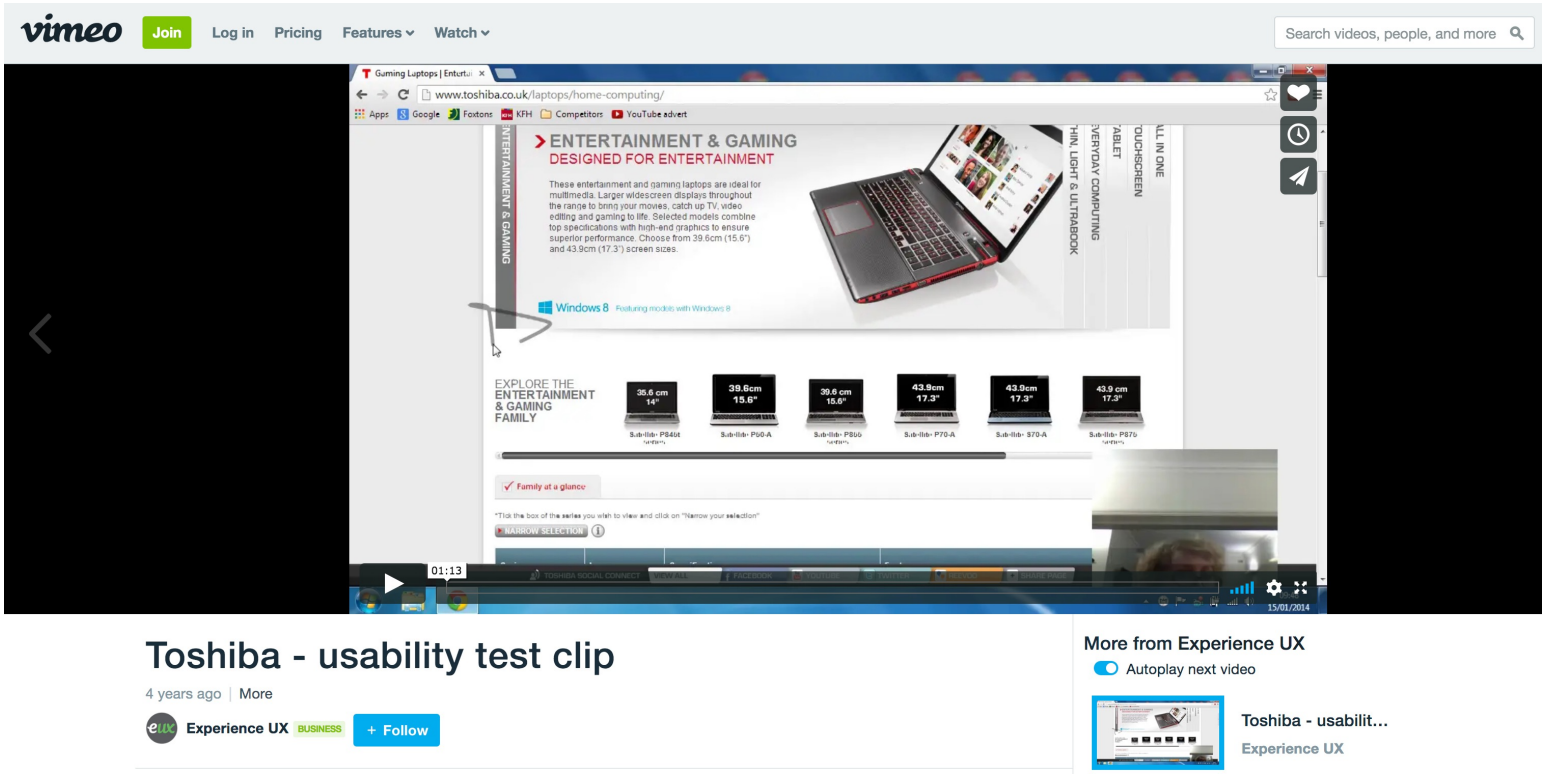
- Users are asked to **speak their thoughts out** loud while doing the task. By speaking they provide information about:
  - What they are trying to do
  - Why they took an action
  - How they interpret what the system did

- Gives insight into the user's mental model of the system, but
  - Risks to be unnatural (awkward and uncomfortable)
  - It is hard to talk if users are concentrating

Video of usability testing of a hi-fi prototype

https://vimeo.com/114778650

- This test involves the Think Aloud technique



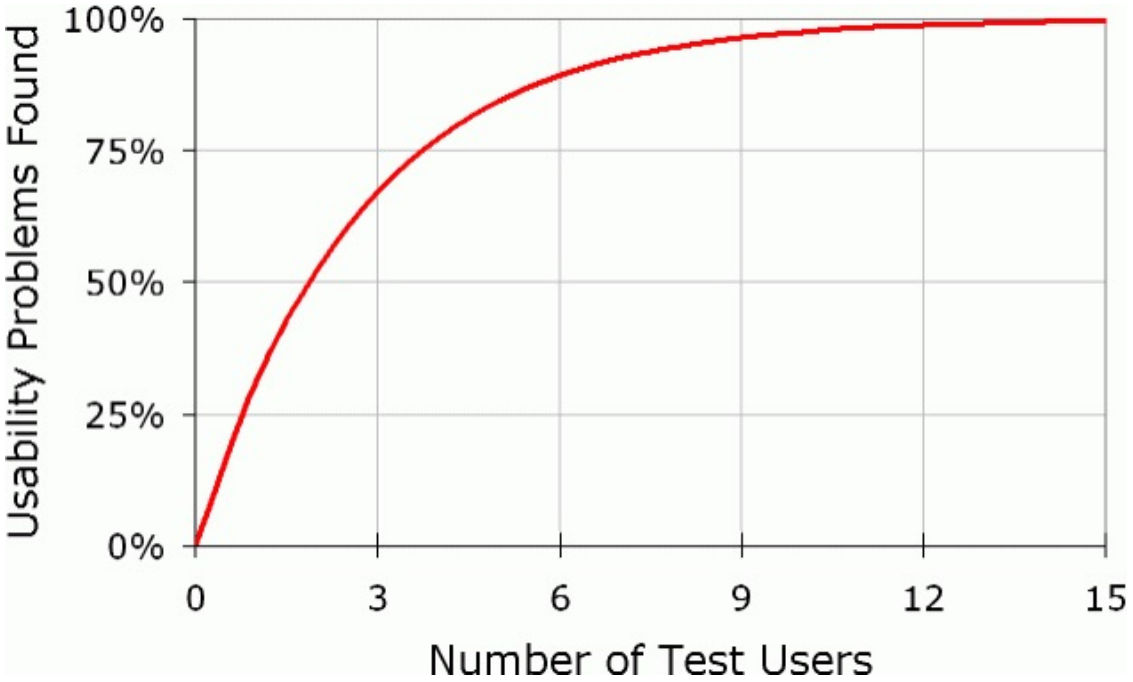
The Wizard of Oz technique



- Some aspects of the interface are implemented (but not all)
- Operation requires processing that is actually done by a human not visible to the user
- The "wizard" intervention **needs to be believable**

You only need to test with 5 users

- According to Nielsen, elaborate usability tests are a waste of resources
- The best results come from testing no more than 5 users and running as **many small tests** as you can afford



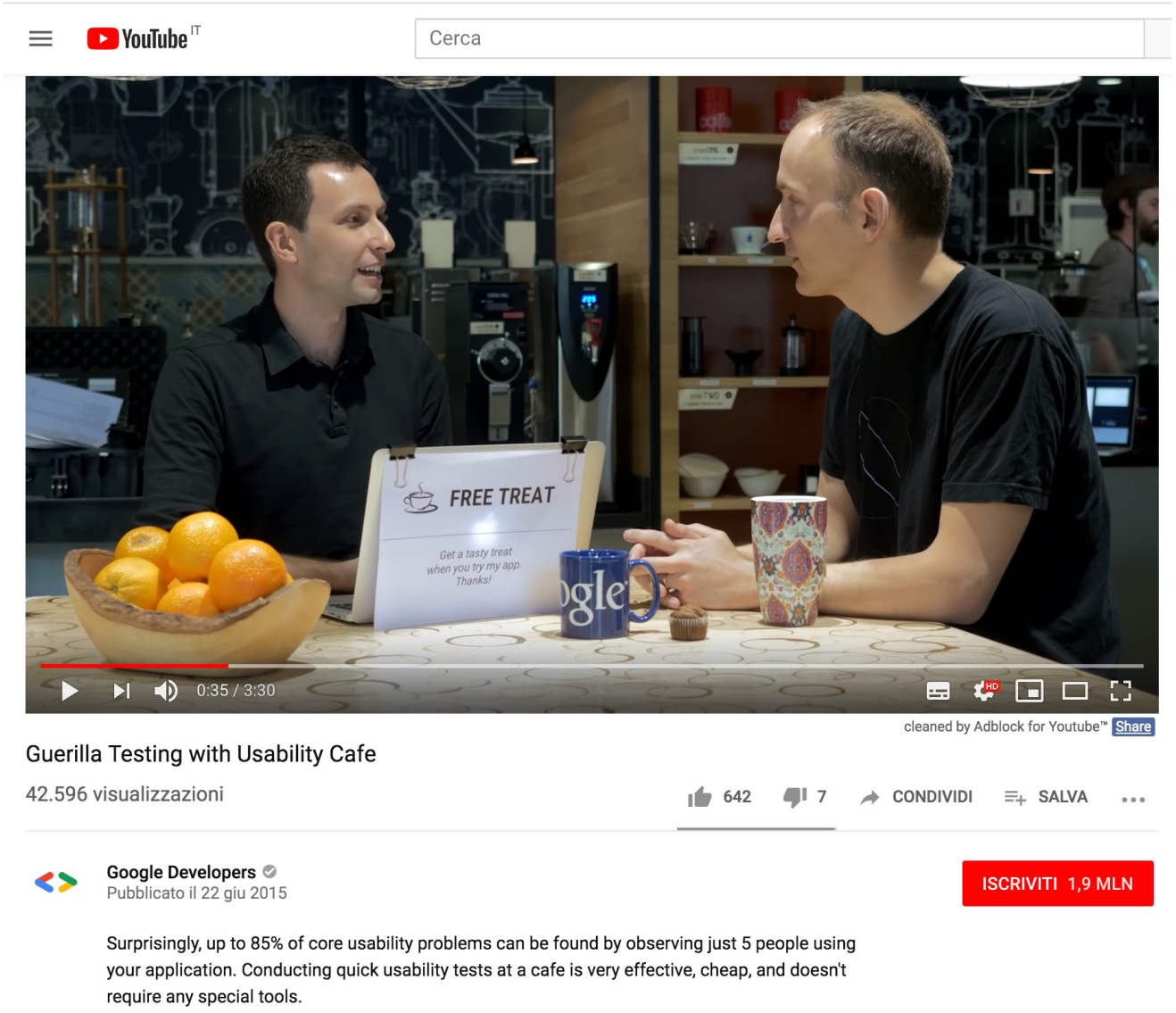
As you add more and more users, you learn less and less because you will keep seeing the same things again and again

Usability testing and iterative design

- You want to run multiple tests because the real goal of usability engineering is to improve the design
- After the first study with 5 participants has found 85% of the usability problems of that specific design, you will want to fix these problems in a redesign
- **After creating the new design, you need to test again**
- Exam: you will have to do 2 iterations

Video of usability testing in action

https://www.youtube.com/watch?v=0YL0xoSmyZI



Note that you test the technology, not the user!

Questionnaire

- Remember that you can also complement the usability testing with a questionnaire at the end of the test
- For instance you could provide the users with the SUS questionnaire

The basics of recruiting users for usability testing

- Recruiting the **right participants** is crucial for getting valuable results from your user research
- Your research participants must be able to represent your target group (i.e., **should possess the characteristics of the users**); otherwise, your results will not translate into something you can use
- Before you recruit participants, you must consider what **criteria** you need to apply when recruiting so that your participants can provide relevant insights for your project
  - E.g., age, expertise in a domain, gender, technological proficiency, etc.

The basics of recruiting users for usability testing

- Deciding on the right number and type of criteria for research participants requires some **tradeoffs**
  - You don't want so few criteria that your participants don't represent your end users
  - You don't want to add criteria that might make it difficult to recruit participants
  - The more specific the criteria you have, the more effort you must put into recruiting, so you should carefully consider the necessity of each criterion
  - Users can have characteristics that are as broad or as narrow as the nature of your project; one thing you should never overlook, though, is **accessibility**: ensure you consider how users with disabilities would encounter your design

The basics of recruiting users for usability testing

- Using your network, or posting on social media can be efficient if your participants are people not too far from your own social circle
- **Be aware of biases (it is highly discouraged to involve friends)**
  - you have less risk of bias if your participants don't know you personally
- If you need to recruit user groups that are hard to reach, you can use interest groups (e.g., meetup.com) or **internet forums** where you think people from your target group might be active (ask permission to moderators)

The basics of recruiting users for usability testing

- It's always a good idea to recruit a few more participants than you actually need; that way, you are sure to have the participants you need in case of no-shows or cancelations
- You can make recruiting easier by making sure that you make the **time and location** of your sessions as convenient as possible for your potential participants and by offering some form of **reward or incentive**
  - The chosen research participants will usually receive some form of reward or financial compensation for their involvement in the research sessions

Field studies

- Field studies are done in natural settings
- **"In the wild"** is a term for prototypes being used freely in natural settings
- Aim to understand what users do naturally and how technology impacts them
- Field studies are used in product design to:
  - Identify opportunities for new technology
  - Determine design requirements
  - Decide how best to introduce new technology
  - Evaluate technology in use

Inspection methods

- The evaluation methods described so far have involved interaction with, or direct observation of, users
- **Inspection methods do not require users to be present during the evaluation**
- They are based on understanding users through one of the following:
  - **Knowledge codified in heuristics**
  - **Data collected remotely (e.g., logging, web analytics)**

Inspection methods based on knowledge codified in heuristics

- Involve an **expert**, role-playing the users for whom the product is designed, analyzing aspects of an interface, and identifying potential usability problems

- The most well-known methods are **Heuristic evaluation**
- For details see the previous lesson about heuristics

Nielsen's 10 heuristics

- **Visibility of system status**
- **Match between system and real world**
- **User control and freedom**
- **Consistency and standards**
- **Error prevention**
- **Recognition rather than recall**
- **Flexibility and efficiency of use**
- **Aesthetic and minimalist design**
- **Help users recognize, diagnose, recover from errors**
- **Help and documentation**

Doing heuristic evaluation

- Briefing session to tell experts what to do
- Evaluation period of 1-2 hours in which:
  - Each expert works separately
  - Take one pass to get a feel for the product
  - Take a second pass to focus on specific features
- Debriefing session in which experts work together to prioritize problems

- In the context of the exam of this course you are the "experts": do the heuristic evaluation

Evaluating for accessibility using guidelines

- Guidelines can be used as heuristics for evaluating websites
- **Web Content Accessibility Guidelines (WCAG)**
  - https://www.w3.org/WAI/standards-guidelines/wcag/
- Four key concepts:
  - **Perceivable**
  - **Operable**
  - **Understandable**
  - **Robust**
  - https://www.w3.org/WAI/standards-guidelines/wcag/glance/

Evaluating for accessibility using guidelines

- **Perceivable**
  - Provide text alternatives for non-text content
  - Provide captions and other alternatives for multimedia
  - Create content that can be presented in different ways, including by assistive technologies, without losing meaning
  - Make it easier for users to see and hear content
- **Operable**
  - Make all functionality available from a keyboard
  - Give users enough time to read and use content
  - Do not use content that causes crisis or physical reactions
  - Help users navigate and find content
  - Make it easier to use inputs other than keyboard

Evaluating for accessibility using guidelines

- **Understandable**
  - Make text readable and understandable
  - Make content appear and operate in predictable ways
  - Help users avoid and correct mistakes
- **Robust**
  - Maximize compatibility with current and future user tools

Which evaluation techniques?

- Depends on:
  - The stage of the design process you are
  - What you want to evaluate (the idea, the usability of the system, etc.)
    - Formulate a research question about what you would like to find
  - The amount of time you have and the level of detail you want to get

- Notes for the exam:
  - Do heuristic evaluation and tests with users (and say it in the report!)
  - You are encouraged to combine different techniques to evaluate your system with users (e.g., usability testing + questionnaires)

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Axure RP (Part 3)

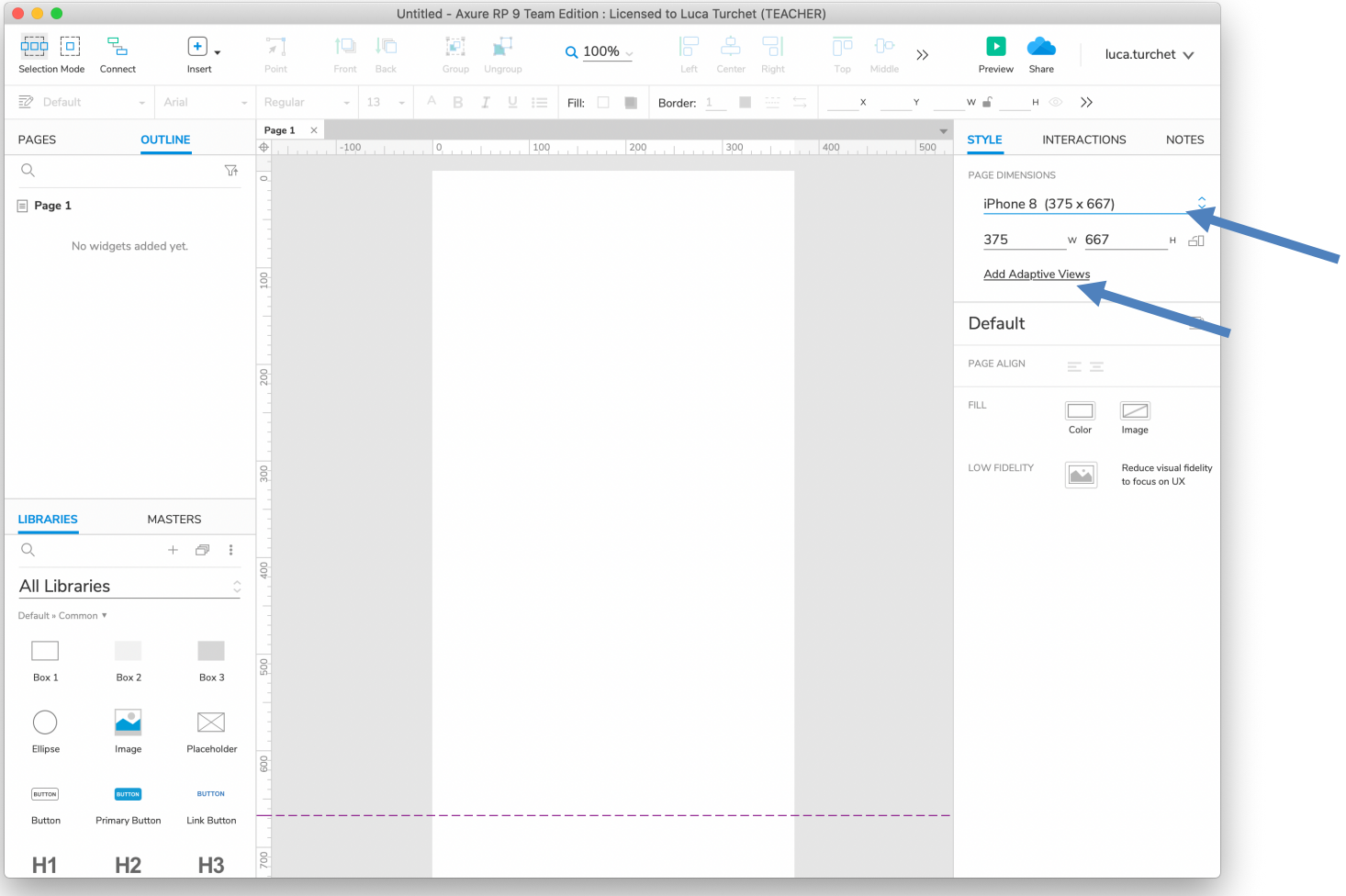
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Hotspots

- https://docs.axure.com/axure-rp/widgets/hot-spots/
- Hot spots are special widgets that are always **invisible** in the web browser
- Use one when you need **to add interactivity to an area of your design where there are no other suitable widgets**, such as when adding a hot zone to a particular area of an image or of a button
- Exercise: 1) create a 2 pages; 2) place a box in each page, with two different colors for each page; 3) add on the top left corner of each box an hot spot; 4) add an interaction to open the other page when clicked on that zone; 5) test the result using the preview.

Exam: set the layout for your project

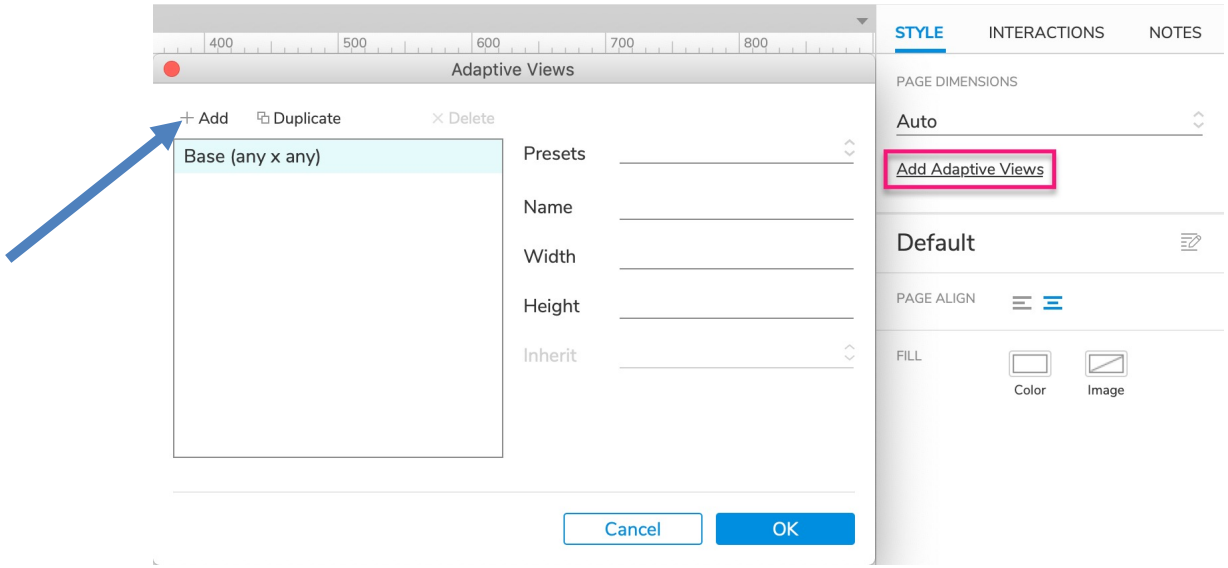


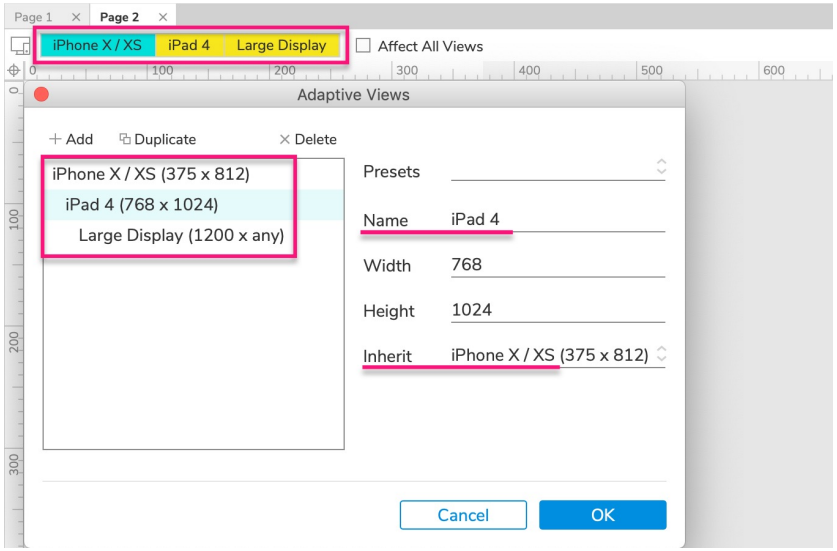
- To test interactions create an adaptive view for your mobile (tablet/smartphone)
- **For the exam: use the preset for iPhone 12 or iPad 4**

Responsive web design: adaptive views

- Are alternate versions of pages designed for specific screen sizes. They allow you to create a page **once** and then rearrange, resize, and restyle its widgets **to fit each screen size** you intend to test it on. When you view the page in a web browser, the appropriate view for the browser dimensions will be shown automatically.
  - https://docs.axure.com/axure-rp/pages/adaptive-views/
  - **Responsive Web design:** with adaptive views you can create responsive prototypes

Adaptive views



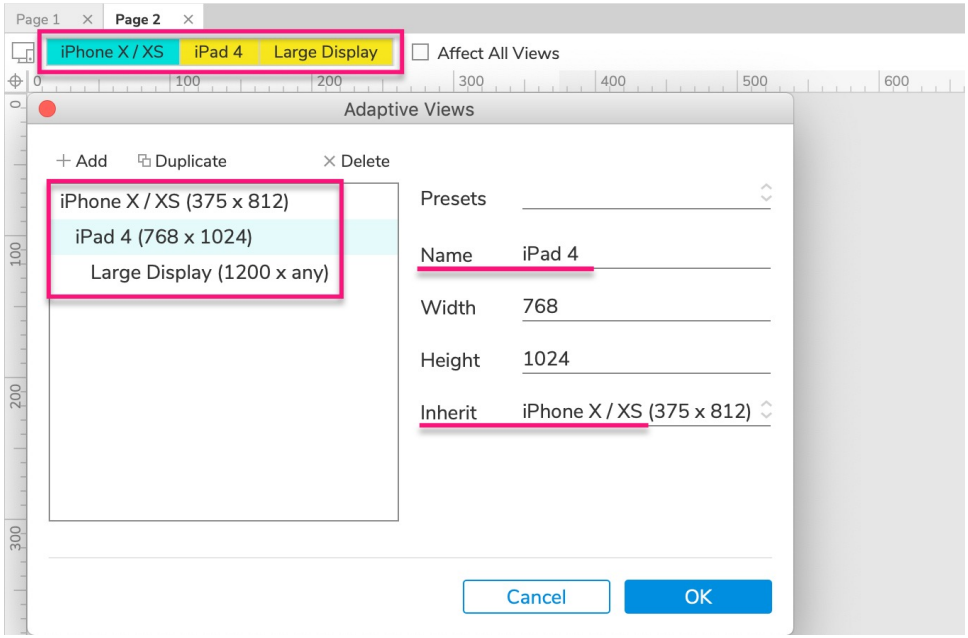


Adaptive views

- For the exam it is enough that you deliver a view for my smartphone so I can test your prototype on my smartphone:
  - Use the preset for iPhone 12: width = 428, height = 926
- You may need to create an adaptive view for your own smartphone so to test the interactions
- However, you can test everything on the web browser on your pc
- Also, for the tests with users you can use the browser so to record the interactions via screen recording
- You can also create an app **and** a website if you like, but this is not mandatory

Adaptive views

- Adaptive view sets contain a series of views organized into a chain of **inheritance**
  - Edits made in the parent's view would be reflected in the children's views as well



iPhone X / XS (Base) > iPad 4 > Large Display

- Edits made in the **iPhone** view would be reflected in both the **iPad** and **Large Display** views as well.
- Edits made in the **iPad** view would be reflected in the **Large Display** view but not in the **iPhone** view.
- Edits made in the **Large Display** view would only affect that view.

Adaptive views

- **Tip:** To take full advantage of the benefits of adaptive view edit inheritance, take a top-down approach to editing your diagrams, starting in the **Base** view and then working your way down the chain
- Ideally, for your project prototype you will need to create various views to fit various smartphones (if your project is an app) **so you can test it with various users even remotely** (thanks to Axure sharing features)

Adaptive views

- **When you view an adaptive project in the web browser, you will automatically be shown the largest view that fits within your browser's width**
  - Use the Adaptive Views dropdown in the prototype player to see the other views
- Sometimes you don't want all widgets of a parent displayed in the children (then just delete them from the children)
  - E.g., you want an horizontal menu in a desktop view but you want a dropdown menu in a mobile view
- If you need a **non-adaptive widget property** to vary across your adaptive views, **create an additional copy** of the widget for each variation and use the *"unplace"* feature to choose which version of the widget appears in each view

Tutorial

- Practice with the following examples in the tutorial:
  - https://docs.axure.com/tutorials/basics/anchor-link/
  - https://docs.axure.com/tutorials/basics/full-width-image-banner/
  - https://docs.axure.com/tutorials/basics/navigation-menu/
  - https://docs.axure.com/tutorials/basics/tab-menu/
  - https://docs.axure.com/tutorials/basics/accordion-menu/

Homework

- Select a website, an app, and a program and assess their usability, by defining a set of tasks and using yourself the Think Aloud technique

Bibliography

- Chapters 14, 15 and 16 of the book "Interaction Design: beyond human-computer interaction" - 5th Edition, by Sharp, Rogers, and Preece. Wiley, 2019.
- https://www.interaction-design.org/literature/topics/usability-testing
- https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/
- https://www.interaction-design.org/literature/article/the-basics-of-recruiting-users-for-usability-testing